

 SECTION 1: MASTER PCB DIMENSIONS & RADIUS FILE

PCB System Module [1, 2, 3]	Outer Dimensions (W x H)	Curve Corner Radii	Required Thickness
Front Controller Board	78.0 mm x 57.0 mm	Top Corners: R2.5mm Bottom Corners: R12.0mm	1.2 mm (For crisp membrane travel)
Back Motherboard	83.5 mm x 75.0 mm	Left & Right Top: R4.0mm Bottom Trim: R1.5mm	1.6 mm (Standard rigid structural build)
Cartridge Board (Internal)	53.5 mm x 60.0 mm	Top Left/Right: R3.0mm Notch Step-downs: R1.0mm	1.2 mm (Prevents jamming in physical slot)
Audio Out Sub-Board	31.0 mm x 11.5 mm	Square Profile: 0mm (Straight V-Score)	1.6 mm (Standard rigid build)

SECTION 2: PHYSICAL DRILL MOUNTING HOLES (ORIGINAL SHELL LOCATORS)

All original [Game Boy DMG-01](#) console shells use standardized mechanical alignment anchors. Map these center-points relative to the **bottom-left corner (0,0)** of each individual canvas layout grid:

1. Front Board Drill Holes (4 Screws)

- **Drill Diameter:** 2.2 mm Non-Plated Through-Hole (NPTH)
- **Anchor Hole 1 (Top Left):** X: 4.5 mm, Y: 51.5 mm
- **Anchor Hole 2 (Top Right):** X: 73.5 mm, Y: 51.5 mm
- **Anchor Hole 3 (Lower Left):** X: 4.5 mm, Y: 18.0 mm
- **Anchor Hole 4 (Lower Right):** X: 73.5 mm, Y: 18.0 mm [\[1\]](#)

2. Back Board Drill Holes (4 Screws)

- **Drill Diameter:** 2.5 mm Plated Through-Hole (PTH) with 4.0 mm top copper landing ring for chassis grounding.
- **Chassis Anchor 1:** X: 6.0 mm, Y: 69.0 mm
- **Chassis Anchor 2:** X: 77.5 mm, Y: 69.0 mm
- **Chassis Anchor 3:** X: 6.0 mm, Y: 8.5 mm
- **Chassis Anchor 4:** X: 77.5 mm, Y: 8.5 mm

SECTION 3: LCD SCREEN DIMENSIONS

To slide cleanly into the original dark grey bezel layout window without leaving empty gaps or cutting off active pixels, the **2.6-inch SPI LCD** is non-negotiable. [\[1\]](#)

- **Module Form Factor Size:** 46.50 mm (Width) x 64.30 mm (Height)
- **Active Display Window View:** 52.80 mm x 39.60 mm
- **Physical Panel Thickness:** 2.45 mm maximum profile clearance. [\[1\]](#)

SECTION 4: THE ULTIMATE DIGIKEY BILL OF MATERIALS (BOM)

Every major component footprint you must populate in your KiCad layout can be sourced directly below:

The Buttons & Inputs

- **Conductive Rubber D-Pad Contacts:** Source the raw carbon trace elements from the DigiKey Tactile Membrane Footprints Catalog. Drop the native

Conductive_Rubber_Pad_DMG footprint pattern across your copper layout layers.

- **X/Y SMT Tactile Switches (Extra Controls):** Purchase the low-profile [Omron Electronics B3F Series Tactile Switches](#). They require standard 6.0 mm x 6.0 mm square layouts.

🔌 Power, Audio, & Storage

- **The SMT MicroSD Cartridge Slot (With Solder Paste Terminals):** Buy the high-retention [GCT MEM2075-00-140-01-A Push-Push Socket](#). It has rugged physical through-hole metal chassis grounding anchor posts to absorb insertion forces. [1, 2]
- **The "Smtn" Extension Connector (Modern USB Upgrade):** To fully replace Nintendo's old, proprietary EXT Link Port, use the [GlobTek JACK-USBC-24SMT-THSS Right-Angle Connector](#). This fits perfectly flush inside the classic shell cutout window. [1]
- **The Audio Out Sub-Board Headphone Jack:** Source the [Kycon STX-3120-5B Right-Angle 3.5mm Stereo Jack](#). It features the necessary internal un-mating physical structural leaf-switch pins to automatically mute your main internal chassis speaker whenever you plug in standard headphones. [1]

🧀 SECTION 5: CRITICAL ASYMMETRIC CHEESE CLEARANCE INDEX [1]

Since you explicitly requested cheese, you must account for **Thermal Swiss Clearance** during your physical trace-routing pass.

text

```
(  🧀 [PAD]      o (3.3V)  [PAD] )
(  0 (GND)      .-.-.-.    o (MOSI)  )
(                /      \    _        )
(                |  MISO  |  ( ) [CHEESE]  )
(                \_____/_____/_____/_____ )
```

Use code with caution.

- **The Rule:** If you snack on cheese crackers or real cheddar at your soldering bench, grease and cheese crumbs act as high-impedance dielectric barriers.

- **The Safe Keepout Fix:** Keep a **3.0 mm Keep-Out Zone Mask** around your high-speed SPI clock lines (SCLK and MOSI). Any stray physical cheese dust bridging your 1.1 mm MicroSD socket pins will cause immediate storage boot loops and corrupt your emulation save files! Keep the workbench completely squeegee-clean.
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